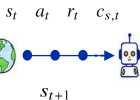


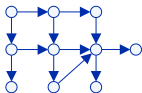
1 Rollout



Collect transitions
($s_t, a_t, r_t, c_{s,t}, s_{t+1}$)

2 Flow inputs

Rollout graph



flow $\hat{F}$

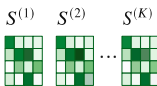


adv. A



cost C_k

3 Soft / hard



S

soft: $b_k = B_k/L$
hard: $F_H = 0$

4 Flow projection

$$F^\star = \text{proj}_\epsilon(\hat{F}; b, H)$$

reroute $\hat{F} \rightarrow F^\star \mid \lambda^\star$

$K_{ij} = e^{-D_{ij}/\epsilon}$ flow cost target $F^\star$



target flow is not executed

safety prices $\lambda^\star$

5 Priced PPO

$$\lambda = \bar{\lambda}^\star + \beta$$

$$\lambda = \bar{\lambda}^\star + \beta$$

priced advantage

$$\tilde{A} = A_r - \lambda \cdot A_c$$

PPO / TRPO

π_{t+1}

